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### **Tunnel FET Analysis Using TCAD Sentaurus**

#### **AIM**

To design, simulate, and analyze a Tunnel Field-Effect Transistor (TFET) using Synopsys TCAD Sentaurus, and to study the effect of gate metal work function, doping concentration, and channel length scaling (50 nm  $\rightarrow$  20 nm) on Ion/Ioff characteristics.

#### **OBJECTIVES**

- To simulate I–V characteristics for different gate metals.
- To analyze gate metal work-function effects.
- To study channel length scaling on TFET behaviour.
- To observe concentration effects on Ion/Ioff.
- To compare device performance using obtained simulations.

#### **Tool:**

**TCAD Sentaurus**

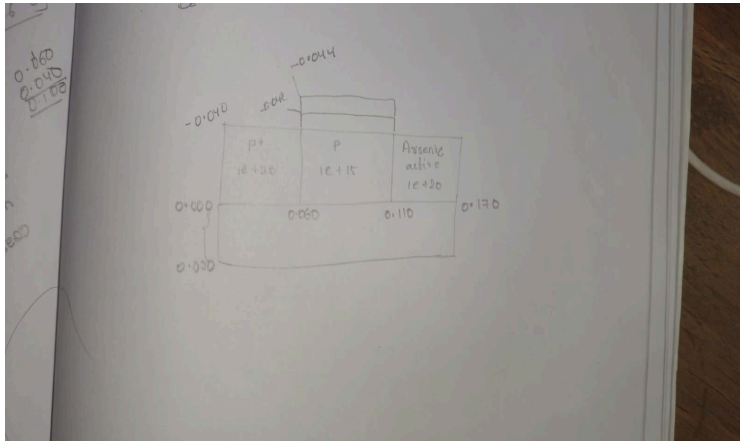
#### **THEORY**

Tunnel Field-Effect Transistors (TFETs) are steep-slope devices that operate on the principle of band-to-band tunneling (BTBT) rather than thermionic emission as in MOSFETs. In a MOSFET, charge carriers require enough thermal energy to overcome the source–channel potential barrier, which naturally limits the subthreshold swing to 60 mV/dec at room temperature. Because TFETs use quantum tunneling across a p–i–n junction instead of thermionic emission, the OFF current can be pushed very low ( $10^{-15}$ – $10^{-18}$  A), and the subthreshold slope can fall well below the 60 mV/dec limit. This makes TFETs excellent candidates for ultra-low-power and energy-efficient applications, especially where leakage control is critical.

The TFET structure typically resembles a lateral p–i–n diode controlled by a gate. For an n-type TFET, the source is heavily p-doped, the channel is intrinsic or lightly doped, and the drain is n+. When the gate voltage is zero, the energy bands at the source–channel junction are far apart. The valence band of the p+ source lies well below the conduction band of the channel. This creates a wide and thick tunneling barrier, resulting in nearly zero tunneling probability. As a result, the TFET remains in a very low leakage OFF state. This is one of the biggest advantages over MOSFETs, which always suffer from thermally generated subthreshold leakage.

As the gate voltage increases, the gate's electric field pulls the channel's energy bands downward. This band bending reduces the energy separation between the source valence band and channel conduction band. When the bands come close enough and the tunneling distance becomes thin, electrons can directly tunnel from the valence band of the source into the conduction band of the channel. This is the turn-on mechanism of a TFET. The moment this alignment improves, the tunneling probability increases very sharply because BTBT depends exponentially on the electric field and barrier width. Hence, the drain current rises extremely quickly with small changes in gate voltage, giving TFETs a steep subthreshold characteristic that is much sharper than MOSFETs.

#### DEVICE STRUCTURE



Source = 60 nm

Channel = 50 nm

Drain = 60 nm

Oxide thickness = 20 nm

Gate metal = 20 nm

BOX thickness = 20 nm

Doping: Source/Drain =  $1e20 \text{ cm}^{-3}$  (As), Channel =  $1e15 \text{ cm}^{-3}$  (Boron)

## PROCEDURE

Create a new project folder in your working directory and open the terminal inside it.

Launch the Sentaurus Structure Editor by typing the command:

Sde

In the Structure Editor, go to View → GUI Mode → 2D Mode to switch the environment to two-dimensional device construction suitable for TFET.

Open the Draw menu, disable Auto Region Naming, and enable Exact Coordinates so that the source, channel, and drain regions can be drawn with the required dimensions:

- Source = 60 nm
- Channel = 50 nm
- Drain = 60 nm

From the Materials list, select Silicon and draw the three rectangular regions corresponding to the source, channel, and drain. After placing each region, name them appropriately (e.g., “source”, “channel”, “drain”).

Apply doping profiles to the silicon regions using

Device → Constant Profile Placement:

- Source ( $p^+$  region): Boron, Active Concentration =  $1 \times 10^{20} \text{ cm}^{-3}$
  - Channel (intrinsic/light p-type): Boron, Active Concentration =  $1 \times 10^{15} \text{ cm}^{-3}$
  - Drain ( $n^+$  region): Arsenic, Active Concentration =  $1 \times 10^{20} \text{ cm}^{-3}$
- This forms the lateral p–i–n junction responsible for band-to-band tunneling in TFET.

Deposit the gate oxide layer by selecting  $\text{HfO}_2$  and drawing a rectangular region of thickness 20 nm above the channel. Name this region “oxide”.

Create the gate metal region on top of the oxide using the selected gate materials (Al, TiN, W). Draw this as a rectangular sheet of thickness 20 nm and name it “gate”.

Draw the BOX (Buried Oxide) layer beneath the silicon body using  $\text{SiO}_2$  with a thickness of 20 nm to complete the SOI TFET structure.

Define the electrical contacts using Contacts → Contact Sets.

Create and name the contact sets for:

- Gate
- Source
- Drain

Change selection mode to Select Edge and apply the contacts to the correct edges of the source, drain, and gate using Contacts → Set Contact.

Define the mesh refinement window using Mesh → Define Rel/Eval Window, and draw a rectangle around the entire device. This ensures accurate meshing, especially near the source–channel tunneling junction.

Apply mesh refinement by going to Mesh → Refinement Placement.

Enter fine minimum element sizes near the source–channel interface because tunneling current requires very high spatial resolution for accurate BTBT modeling.

Build the final mesh using Mesh → Build Mesh.

The generated mesh file (.tdr) will open automatically in svisual for inspection.

Run the simulation using the terminal command:

`sdevice project.cmd`

## RESULTS AND FIGURES

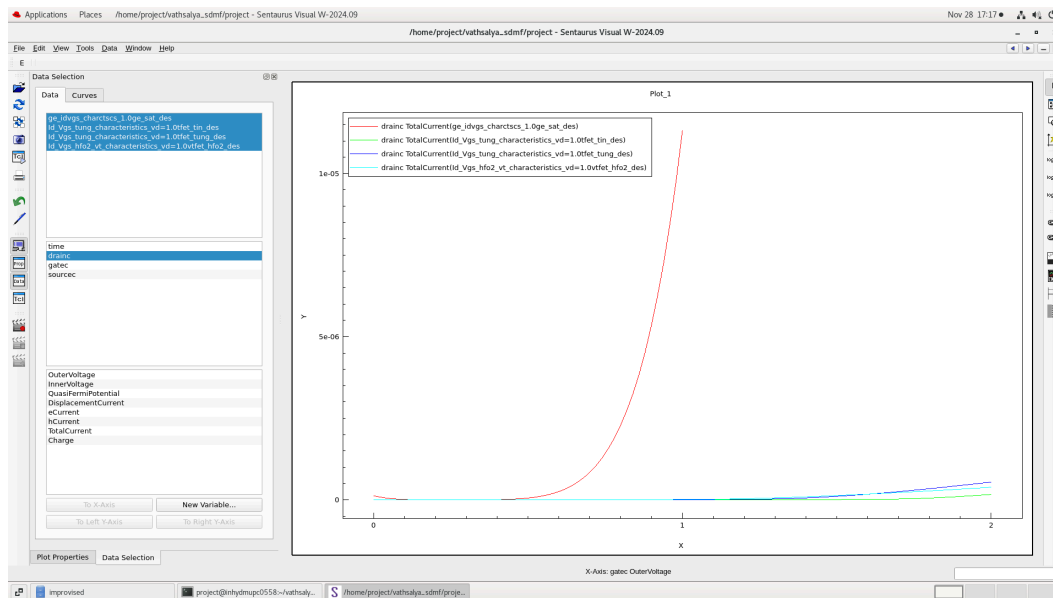


Figure: 50nm\_id\_vs\_vgs\_metal\_ge

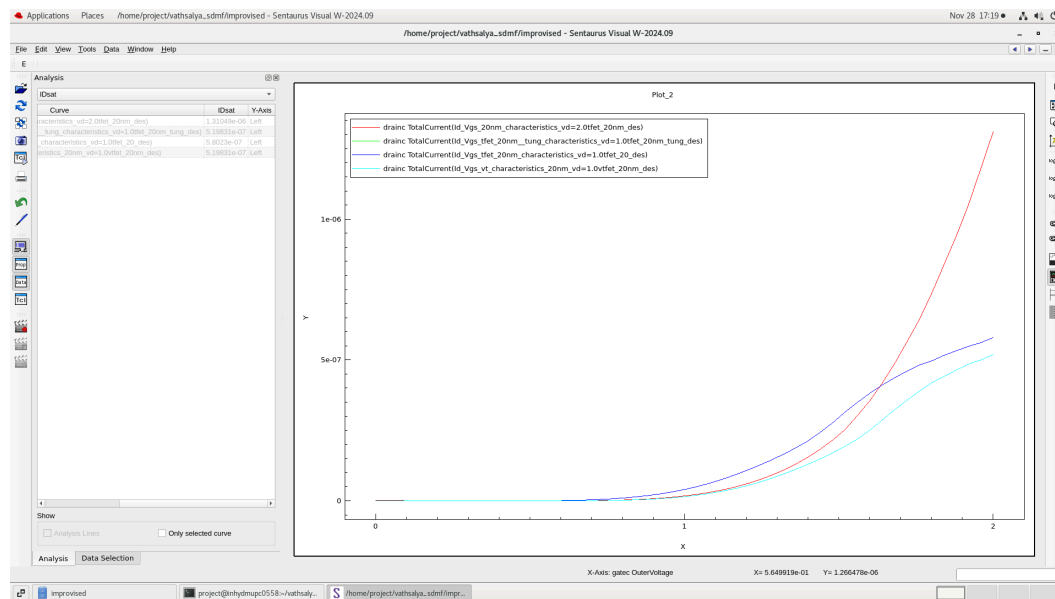


Figure: id\_on\_20nm

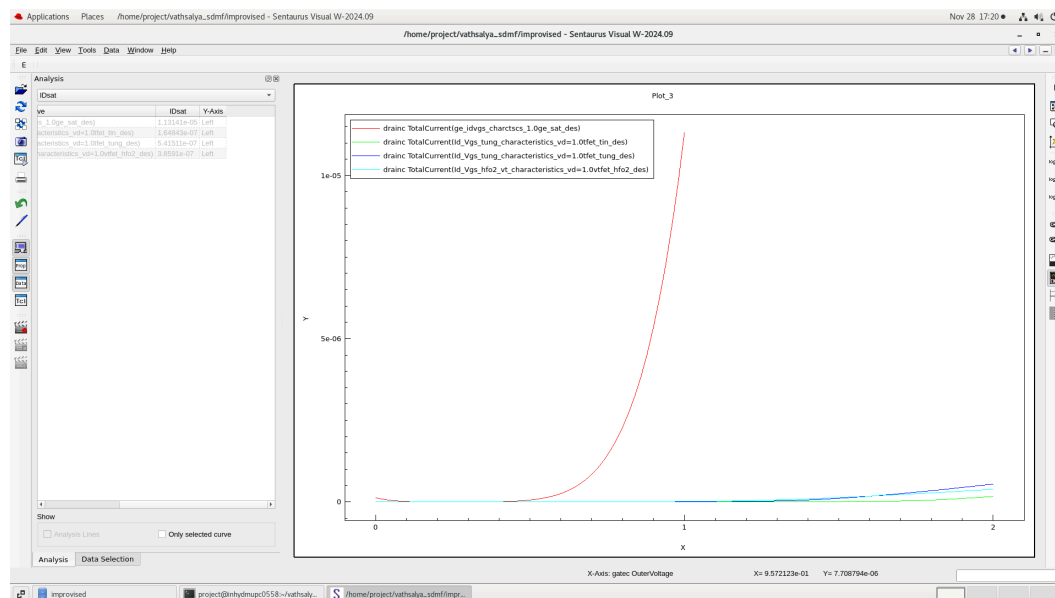


Figure: id\_on\_50nm

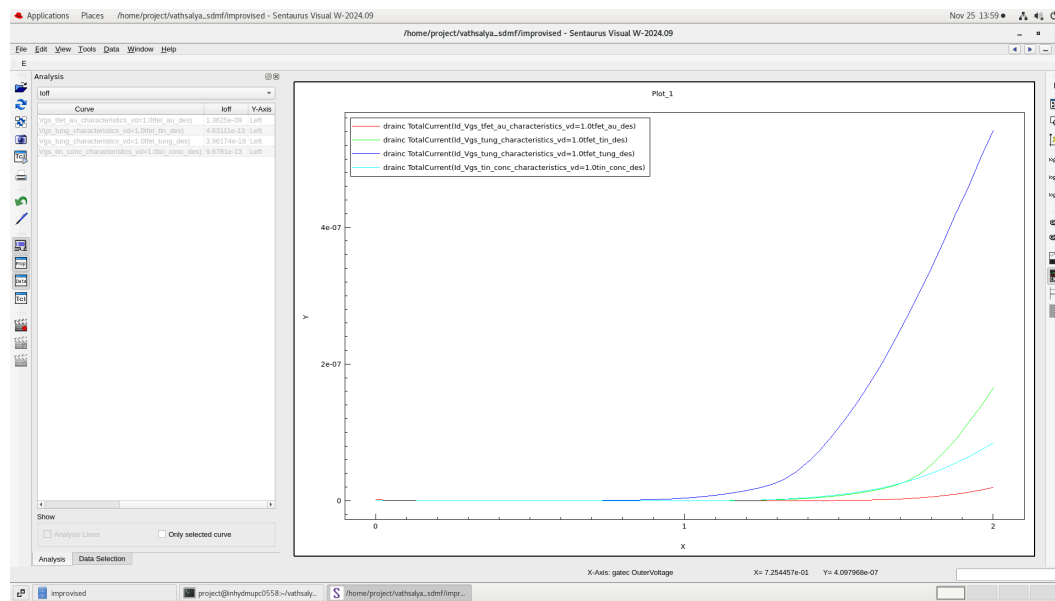


Figure: ioff\_comp

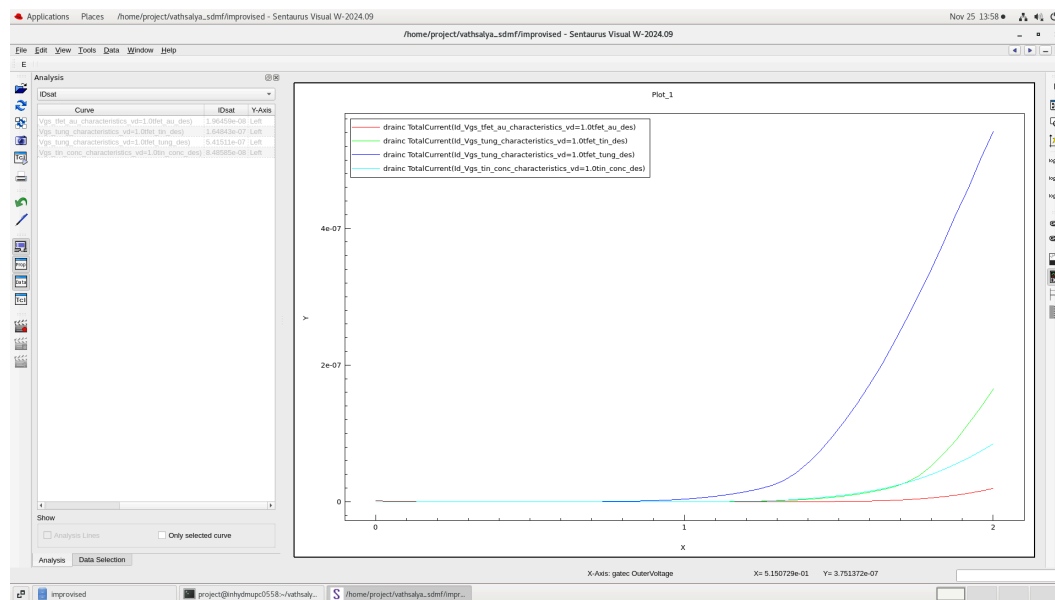


Figure: id\_sat\_comp

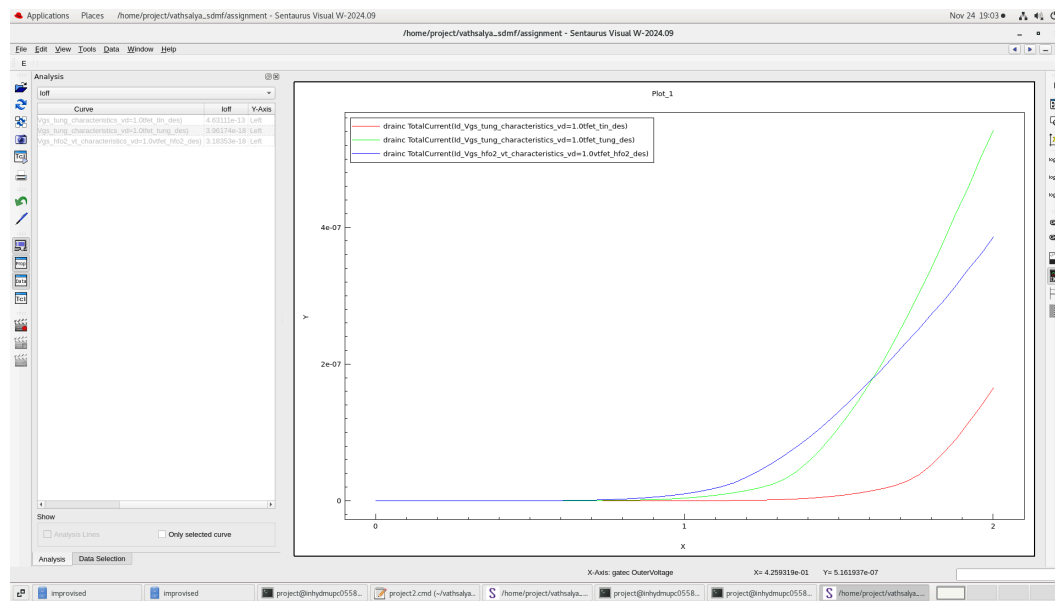


Figure: ioff\_tin\_al\_tung

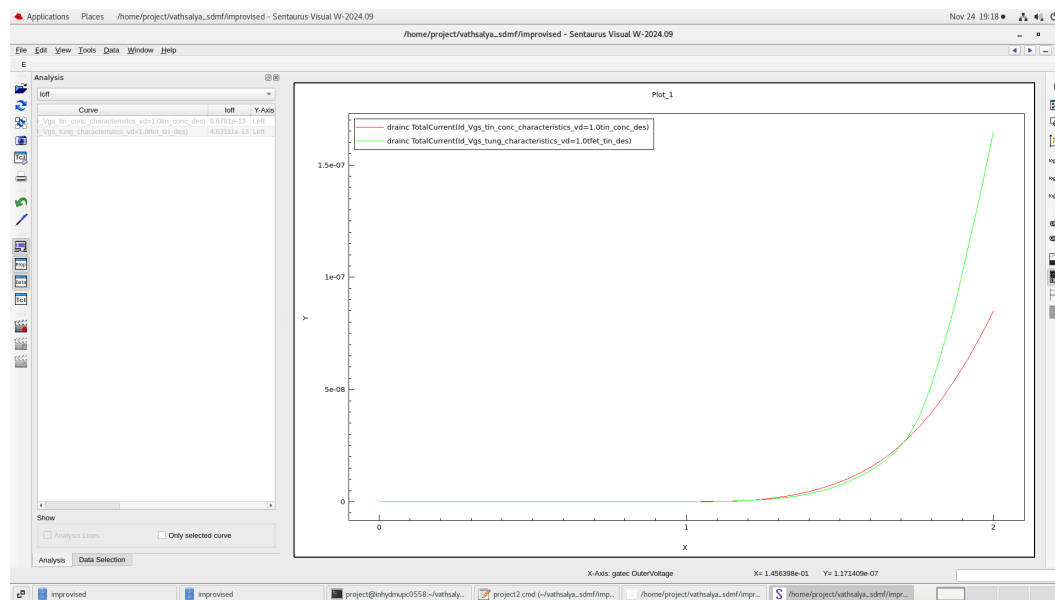


Figure: concn\_ioff

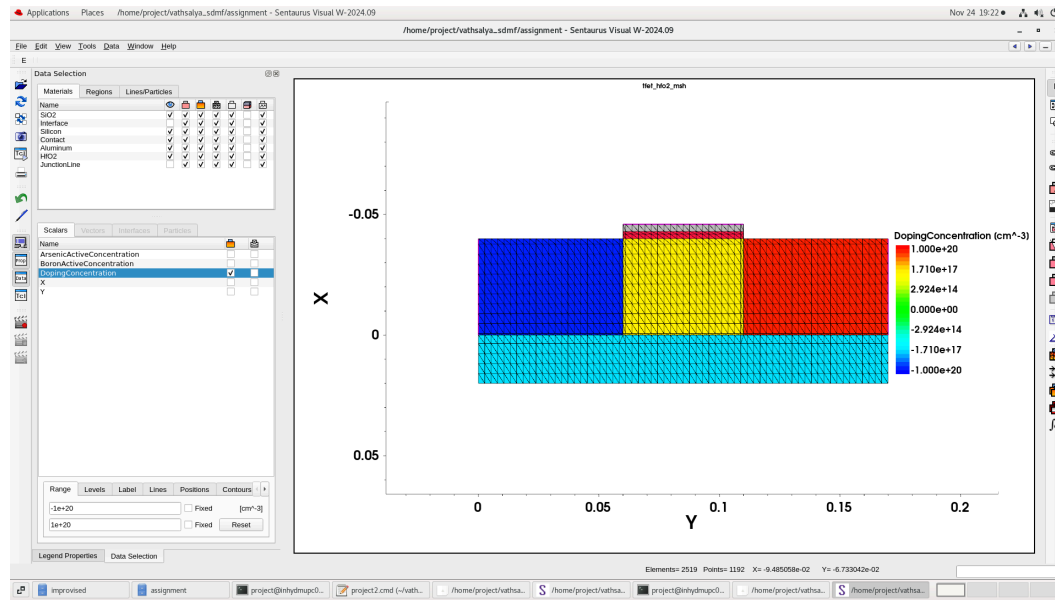


Figure: mesh

## OBSERVATIONS

### 1. Gate Metal Effect

Low work-function metals (like Al) bend the channel bands more → tunneling barrier becomes thinner → Ion increases.

High work-function metals (like TiN, W) cause less band bending → tunneling barrier becomes thicker → Ioff decreases.

TiN achieves the best Ion/Ioff, because it provides a balance between strong enough tunneling in ON state and very low tunneling in OFF state.

### 2. Channel Length Scaling (50 nm → 20 nm)

In 50 nm channel, gate control over the tunneling junction is weaker → electric field is spread out → lower tunneling probability → lower Ion.

In 20 nm channel, gate field is stronger and more concentrated → tunneling width reduces → Ion increases.

Strong field at small channel lengths slightly bends bands even in OFF state → tunneling does not fully stop → Ioff increases slightly.

### 3. Concentration Effect

Increasing source doping increases built-in electric field and reduces depletion width → higher tunneling rate → higher Ion.

Very high doping makes depletion region too narrow even at low VGS → unwanted band overlap → Ioff increases.

Excessive doping causes bandgap narrowing (band tailing), which also increases OFF-state tunneling → Ioff rises exponentially.

## CALCULATIONS:



Best Gate Metal (TiN):

$$I_{on} \approx 4.5 \times 10^{-7} \text{ A}$$

$$I_{off} \approx 4.6 \times 10^{-17} \text{ A}$$

$$I_{on}/I_{off} \approx 9.7 \times 10^{10}$$

Best Channel Length (20 nm):

$$I_{on} \approx 1.3 \times 10^{-6} \text{ A}$$

$$I_{off} \approx 9.6 \times 10^{-13} \text{ A}$$

$$I_{on}/I_{off} \approx 1.3 \times 10^6$$

## CONCLUSION

The TFET simulation confirms that device behavior is highly sensitive to gate metal work function, doping levels, and channel length. A 20 nm channel length improves  $I_{on}$  due to stronger electric fields, while TiN provides the best  $I_{on}/I_{off}$  performance. TFETs show extremely low leakage, validating their suitability for ultra-low-power circuits.